# Ex No: 13 Implement your own ping program

Aim:

To implement a simple ping program in computer networks to check whether a destination host is reachable and to measure the round-trip time (RTT) of packets.

Algorithm:

i) Start the program
ii) Read the destination host name or IP address from the user.
iii) Create a raw ICMP socket.
iv) Construct an ICMP Echo Request packet containing:
    ICMP type
    Code
    Checksum
    Identifier
    sequence number
    Data
Calculate the checksum of the packet. Send the ICMP Echo Request to the destination.
Wait for an ICMP Echo Reply.
Calculate the round-trip time between sending the request and receiving the reply.
Display the destination, packet size, sequence number, and RTT.
Repeat the process for a specified number of packets.
Display the final result and terminate.

Program - Python

```
import socket
import struct
import time
import os
def checksum(data):
    total = 0
    length = len(data)
    for i in range(0, length-1, 2):
        total += (data[i] << 8) + data[i+1]
    if length % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
    total += total >> 16
    return (~total) & 0xffff
def ping(host):
    try:
        dest_addr = socket.gethostbyname(host)
    except socket.gaierror:
        print("Unable to resolve host")
        return
    icmp = socket.getprotobyname("icmp")
    try:
        sock = socket.socket(socket.AF_INET, socket.SOCK_
RAW, icmp)
    except PermissionError:
        print("Permission denied. Run the program
with administrator/root privileges.")
    sock.settimeout(2)
    packet_id = os.getpid() & 0xFFFF
    sequence = 1
    header = struct.pack("!BBHHH", 8, 0, 0, packet_id,
sequence)
    packet = header + data
    start_time = time.time()
```

```
        try:
            sock.sendto(packet, (dest_addr, 1))
            sock.recvfrom(1024)
            end_time = time.time()
            rtt = (end_time - start_time) * 1000
            print("Reply from", dest_addr,
                  ": bytes =", len(packet),
                  "time = %.2f ms" % rtt)
        except socket.timeout:
            print("Request timed out")
        finally:
            sock.close()

# Main program
host = input("Enter hostname or IP address: ")
for i in range(4):
    ping(host)
    time.sleep(1)
```

Sample output:

```
Enter hostname or IP address: 8.8.8.8
Reply from 8.8.8.8 : bytes = 43 time = 24.91 ms
Reply from 8.8.8.8 : bytes = 43 time = 23.87 ms
Reply from 8.8.8.8 : bytes = 43 time = 24.12 ms
Reply from 8.8.8.8 : bytes = 43 time = 24.36 ms
```

Result:

Thus, a ping program was successfully implemented using Python. The program sends ICMP Echo Request packets to the specified hosts, receives Echo Reply packets, and calculates the round-trip time (RTT) to determine network connectivity.